

Fourier Series

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Page No.

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Fourier series is a mathematical tool that allows the representation of any periodic signal in terms of sum of harmonically related sinusoids.

For CT

$$e^{j\omega_0 t + \phi} \rightarrow \boxed{h(t)} \rightarrow y(t) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega_0 \tau} \cdot e^{j\omega_0 t + \phi} d\tau$$
$$\boxed{y(t) = H(\omega_0) \cdot e^{j\omega_0 t + \phi}}$$

* Provided the integral converges

$$y(t) = |H(\omega_0)| \cdot e^{j(\omega_0 t + \phi(\omega_0))}$$

$h(t) \rightarrow$ Impulse Response (system)

$H(\omega) \rightarrow$ Freq. Response

$$\text{Eg.} \rightarrow \sum x_i \cdot e^{j\omega_i t} \rightarrow \sum x_i \cdot H(\omega_i) \cdot e^{j\omega_i t}$$

$$\Rightarrow x(t) \rightarrow \int_{-\infty}^{\infty} h(\tau) \cdot x(t-\tau) d\tau$$

$$\Rightarrow X_i \rightarrow X_i H(\omega_i)$$

For DT

$$e^{j\omega n} \rightarrow \boxed{h[n]} \rightarrow \sum_k h[k] \cdot e^{j\omega k} \cdot e^{j\omega n} = y[n]$$

$$\boxed{y[n] = H(\omega) \cdot e^{j\omega n}}$$

* Provided if summation converges

freq. response
of DT signal

⇒ The LTI systems have a unique property that a response to a complex expo. is the same complex expo. with a complex gain denoted by:

$$\begin{aligned} \text{(i)} \quad H(\omega) &= \int_{-\infty}^{\infty} h(\tau) \cdot e^{-j\omega\tau} \cdot d\tau \\ \text{(ii)} \quad H(\Omega) &= \sum_{k=-\infty}^{\infty} h[k] e^{-j\Omega k} \end{aligned} \quad \left. \vphantom{\begin{aligned} \text{(i)} \quad H(\omega) &= \int_{-\infty}^{\infty} h(\tau) \cdot e^{-j\omega\tau} \cdot d\tau \\ \text{(ii)} \quad H(\Omega) &= \sum_{k=-\infty}^{\infty} h[k] e^{-j\Omega k} \end{aligned}} \right\} \begin{array}{l} \text{Provided integral} \\ \text{or summation} \\ \text{converges} \end{array}$$

→ We know that an LTI system is completely characterised by its impulse response $h(t)$ / $h[n]$. Thus, the same LTI system is also completely characterised by its freq. response $H(\omega)$ / $H(\Omega)$. (provided they converge)

$$\begin{aligned} * \quad e^{j\omega_0 t} &\rightarrow T_0, \frac{2\pi}{\omega_0}, f_0 \\ e^{j\Omega_0 t} &\rightarrow T_0, \frac{2\pi}{\omega_0}, k f_0 \end{aligned} \quad \left\{ \begin{array}{l} \boxed{x(t) = \sum_{k=-\infty}^{\infty} x_k e^{-j\Omega_0 k t}} \\ T_0, k f_0, k \in \mathbb{Z} \end{array} \right.$$

⇒ A periodic signal $x(t)$ with T.P. = T_0 & satisfies the Dirichlet condⁿ then $x(t)$ can be represented as linear-weighted $\downarrow$ of ∞ harmonics
Summation

** C.T. Fourier Series is applicable for all power signals

⇒ This representation is Fourier Series representation of periodic signals and the coeff. x_k 's are known as Fourier Series Coefficients.

$$x[n] = \sum_{k=0}^{N-1} x_k \cdot e^{j\Omega_0 n k} \Rightarrow \text{D.T. Fourier series}$$

↳ ∴ Only N Harmonics

Dirichlet's Condition

- ① Given that $x(t)$ is a periodic signal with T.P. = T_0 then $x(t)$ can be represented as a CFS i.e.,

CFS $\rightsquigarrow$
$$x(t) = \sum_{k=-\infty}^{\infty} X_k \cdot e^{j\omega_0 k t}$$
 , $\omega_0 = \frac{2\pi}{T_0}$

where ω_0 or f_0 is the fundamental freq. and X_k are the Fourier series coeff., provided that $x(t)$ satisfies Dirichlet's condⁿ. :-

- (i) $x(t)$ is absolutely integrable over the time period
- (ii) $x(t)$ has finite maximal minima in one time period.
- (iii) $x(t)$ has finite no. of discontinuities in one T.P.

- ② Given that $x[n]$ is a periodic signal with T.P. = N , then the periodic signal $x[n]$ can be represented as ;

$$x[n] = \sum_{k=0}^{N-1} X_k \cdot e^{j\omega_0 k n} \quad \omega_0 = \frac{2\pi}{N}$$

This is known as DTFS

and X_k : $k=0, 1, 2, \dots$ are F.S. Coeff.

$$(3) (a) \quad \frac{1}{T_0} \int_0^{T_0} x(t) \cdot e^{-j\omega_0 t} dt = \langle x(t), e^{j\omega_0 t} \rangle$$

$$(b) \quad \frac{1}{T_0} \int_0^{T_0} \sum_{k=-\infty}^{\infty} X_k \cdot e^{j\frac{2\pi}{T_0}(k-p)t} dt = \frac{1}{T_0} \sum_{k=-\infty}^{\infty} X_k \underbrace{\int_0^{T_0} e^{j\omega_0(k-p)t} dt}_{T_0 \delta(k-p) \text{ at } k=p}$$

$$= \sum_{k=-\infty}^{\infty} X_k \cdot \frac{T_0}{T_0} \delta(k-p)$$

$$= \underline{X_p} \quad (\text{at } k=p)$$

(c) Therefore, from above 2 eqns ;

$$X_k = \frac{1}{T_0} \int_0^{T_0} x(t) \cdot e^{-j\omega_0 k t} dt \quad \rightarrow \text{F.S. Coeff for C.T.}$$

$$(4) (a) \quad X_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] \cdot e^{-jkn\omega_0} \quad \rightarrow \text{F.S. Coeff for D.T.}$$

$$(b) \quad X_{k+N} = \frac{1}{N} \sum x[n] \cdot e^{-j(k+N)n\omega_0} = \frac{1}{N} \sum x[n] \cdot e^{-jkn\omega_0} \cdot e^{-jNn\omega_0}$$

$$X_{k+N} = X_k$$

Ex. 1) $x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$

Solⁿ $x(t)$ is periodic TP = T_s

$$\Rightarrow X_k = \frac{1}{T_s} \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} \delta(t) \cdot e^{-jk \frac{2\pi}{T_s} t} dt$$

$$= \frac{1}{T_s} \int_{-\frac{T_s}{2}}^{\frac{T_s}{2}} \delta(t) dt = \frac{1}{T_s}$$

(Time shifting property)

Note $\rightarrow$

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_s) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} e^{jk \frac{2\pi}{T_s} t}$$

} Poisson Summation Formula

Ex. 2) $x[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN_0]$

Solⁿ $x[n]$ is periodic with TP = N_0

$$\Rightarrow X_k = \frac{1}{N_0} \sum_{n=0}^{N_0-1} x[n] \cdot e^{-jk \frac{2\pi}{N_0} n}$$

$$= \frac{1}{N_0} \sum_{n=0}^{N_0-1} \delta[n] \cdot e^{-jk \frac{2\pi}{N_0} n}$$

$$= \frac{1}{N_0} \sum_{n=0}^{N_0-1} \delta[n]$$

(Time shifting property)

$$= \frac{1}{N_0}$$

Note

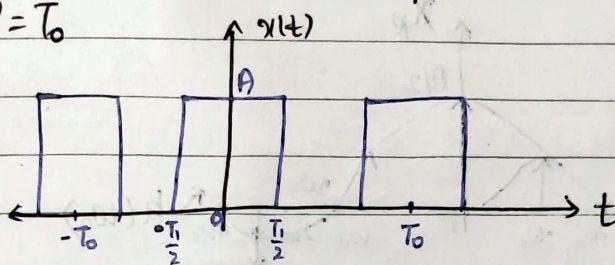
→

$$x[n] = \sum_{k=-\infty}^{\infty} \delta[n - kN_0] = \frac{1}{N_0} \sum_{k=0}^{N_0-1} e^{j \frac{2\pi}{N_0} nk}$$

}

Poisson Summation Formula

Eg: Consider a periodic rect. pulse train with pulse width $= T_1$
 & T.P. $= T_0$



soln

$$X_k = \frac{1}{T_0} \int_{-T_1/2}^{T_1/2} x(t) \cdot e^{-j \frac{2\pi}{T_0} kt} dt = \frac{1}{T_0} \int_{-T_1/2}^{T_1/2} A \cdot e^{-j \frac{2\pi}{T_0} kt} dt$$

$$\Rightarrow X_k = \frac{A}{T_0} \left[\left[-j \frac{2\pi}{T_0} k \right]^{-1} \cdot e^{-j \frac{2\pi}{T_0} kt} \right]_{-T_1/2}^{T_1/2}$$

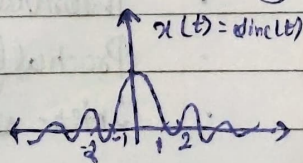
$$\Rightarrow X_k = \frac{-A}{2\pi j k} \left[e^{-j \frac{\pi k T_1}{T_0}} - e^{j \frac{\pi k T_1}{T_0}} \right] = \frac{A T_1}{2\pi k} (-2j) \left[\sin \frac{k \pi T_1}{T_0} \right]$$

$$\Rightarrow X_k = \frac{A}{\pi k} \sin \left(\frac{k T_1 \pi}{T_0} \right)$$

$$\Rightarrow X_k = \frac{A \cdot T_1}{T_0} \frac{\sin \left(\frac{k \pi T_1}{T_0} \right)}{\left(k \pi \cdot \frac{T_1}{T_0} \right)}$$

$$\left[\text{sinc}(x) = \frac{\sin \pi x}{\pi x} \right]$$

$$\Rightarrow X_k = A \cdot \frac{T_1}{T_0} \cdot \text{sinc} \left(\frac{k T_1}{T_0} \right)$$

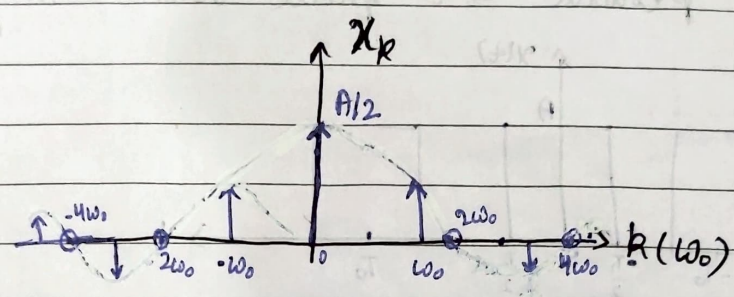


Case ①: T_0 is constant $T_0 = mT_1$

$$m = 2 : \frac{T_1}{T_0} = \frac{1}{2}$$

$$\Rightarrow X_k = \frac{A}{2} \text{sinc}\left(\frac{k}{2}\right)$$

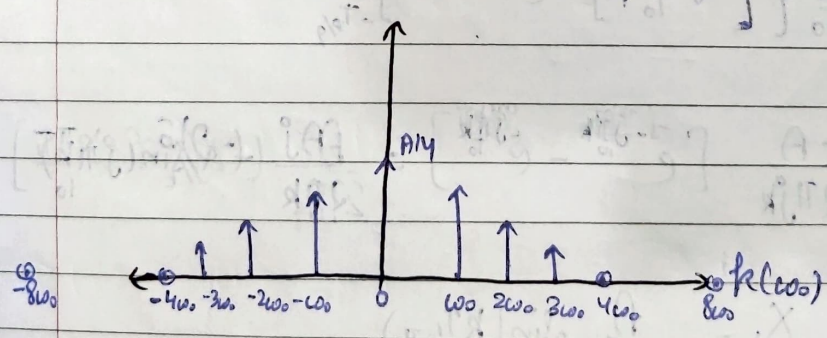
$$B.W = 4\omega_0$$



Case ②: $m = 4 : \frac{T_1}{T_0} = \frac{1}{4} \quad T_0 = mT_1$

$$X_k = \frac{A}{4} \text{sinc}\left(\frac{k}{4}\right)$$

$$B.W = 8\omega_0$$



Point. For a rect. pulse train we observe that for a constant T_1 $\exists$ a varying pulse width.

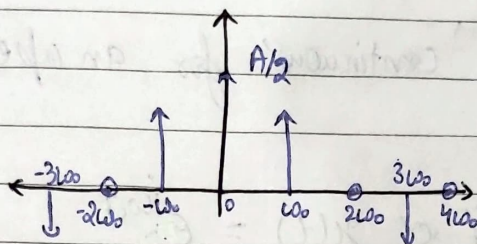
- Theoretically there exists ∞ freq. $\therefore$ Band Width $= \infty$
- Practically, we may consider the null to null B.W
- Inter-spectral spacing is constant $= \omega_0 = \frac{2\pi}{T_0}$
- $B.W \uparrow$ and $T_1 \downarrow$ or $B.W \downarrow$ and $T_1 \uparrow$

Case ③: T_1 is constant, T_0 is varying

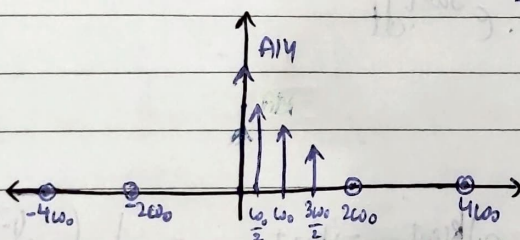
let $T_0 = m T_1$

$$\Rightarrow X_k = \frac{A T_1}{T_0} \sin\left(k \frac{T_1}{T_0}\right)$$

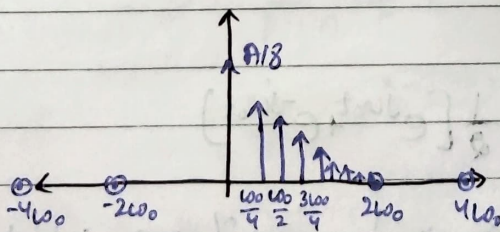
$$\Rightarrow m=2 : T_0 = 2T_1 \Rightarrow \omega_0 = \frac{\pi}{T_1}, X_k = \frac{A}{2} \sin\left(\frac{k}{2}\right)$$



$$\Rightarrow m=4 : T_0 = 4T_1 \Rightarrow \omega_0 = \frac{\pi}{2T_1} = \frac{\omega_0}{2}, X_k = \frac{A}{4} \sin\left(\frac{k}{4}\right)$$



$$\Rightarrow m=8 : T_0 = 8T_1 \Rightarrow \omega_0 = \frac{2\pi}{8T_1} = \frac{\pi}{4T_1} = \frac{\omega_0}{4}, X_k = \frac{A}{8} \sin\left(\frac{k}{8}\right)$$



Freq. of Zero Crossing

$$= 2\omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{T_1}$$

Points

$\Rightarrow$ If pulse width does not change then T.P $\rightarrow \infty$

- Line spectrum $\rightarrow$ Continuous spectrum
 - Zero Crossing $\rightarrow$ Fixed
- $\therefore$ B.W. is fixed or same

- If the pulse width $\downarrow$ BW $\uparrow$
- Line spacing $\downarrow$ as TP $\uparrow$
- Spectrum shall consist of 'lines' for a periodic signal
- " " " 'continuous' for an aperiodic signal

Ex $\rightarrow$ find the F.S. Coeff of $x(t) = e^{j\omega_0 t}$

$$x(t) = \sum_{k=-\infty}^{\infty} X_k \cdot e^{jk\omega_0 t}$$

Soln $\Rightarrow$ $X_1 = \frac{1}{T_0} \int_0^{T_0} e^{j\omega_0 t} \cdot e^{-j\omega_0 t} dt$

$$\underline{X_1 = 1}$$

$$\Rightarrow X_k = \frac{1}{T_0} \int_0^{T_0} 1 \cdot e^{j\omega_0 t} \cdot e^{-jk\omega_0 t} dt = \frac{1}{T_0} \int_0^{T_0} e^{j(1-k)\omega_0 t} dt$$

$$\underline{X_k = 0} \quad (k \neq 1)$$

Ex $\rightarrow$ (9) $x(t) = \cos \omega_0 t = \frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}]$

$$\Rightarrow X_1 = \frac{1}{2}, \quad X_{-1} = \frac{1}{2}, \quad X_k = 0 \quad : |k| > 1$$

[An Integ for Fourier Series Coeff]

(10) $y(t) = \sin \omega_0 t = \frac{1}{2j} [e^{j\omega_0 t} - e^{-j\omega_0 t}]$

$$\Rightarrow Y_1 = \frac{1}{2j}, \quad Y_{-1} = -\frac{1}{2j}, \quad Y_k = 0 \quad (|k| > 1)$$

⇒ Fourier Properties :-

① (a) Consider a real periodic signal; $x(t) = x^*(t)$

$$\sum_k X_k \cdot e^{jk\omega t} = \sum_k X_k^* \cdot e^{-jk\omega t}$$

$$= \sum_p X_{-p}^* \cdot e^{j\omega t p}$$

← let $-k=p$

$$= \sum_k X_{-k}^* \cdot e^{j\omega t k}$$

↙ $p \rightarrow k$

On comparing: $X_k = X_{-k}^*$

$$\Rightarrow \underline{x[k] = x^*[-k]}$$

Discrete freq. sequence
or signal

(b) $x[n]$ is real $\therefore x[n] = x^*[n]$

$$\sum_{k=0}^{N-1} X_k \cdot e^{j\frac{2\pi}{N}nk} = \sum_{k=0}^{N-1} X_k^* \cdot e^{-j\frac{2\pi}{N}nk}$$

let $k=-p$

Similar to above (a) $\hookrightarrow = \sum_{p=1-N}^0 X_{-p}^* \cdot e^{+j\frac{2\pi}{N}np}$

$$X_k = X_{-k}^*$$

$$= \sum_k X_k^* \cdot e^{j\frac{2\pi}{N}nk}$$

$$X_k = X_{-k}^*$$

On Comparing $\Rightarrow X_k = X_{-k}^*$

~~$X_k = X_{k+N}$~~ $\Rightarrow$ p^{th} harmonic is same as $(p+N)^{\text{th}}$

$$\therefore X_{p+N} = X_p \Rightarrow \sum_{p=1-N}^0 X_{N-p}^* \cdot e^{j\frac{2\pi}{N}n(N+p)}$$

$$\Rightarrow X_0 = X_0^*$$

$$X_3 = X_3^* = X_{N-3}^*$$

$$X_1 = X_1^* = X_{N-1}^*$$

$$X_2 = X_2^* = X_{N-2}^*$$

$$X_p = X_{-p}^* = X_{N-p}^*$$

Let $N=7$;

$$\left\{ \begin{array}{l} X_0 = X_0^* = X_7^* \\ X_1 = X_1^* = X_6^* \\ X_2 = X_2^* = X_5^* \\ X_3 = X_3^* = X_4^* \\ X_4 = X_4^* = X_3^* \\ X_5 = X_5^* = X_2^* \\ X_6 = X_6^* = X_1^* \\ X_7 = X_7^* = X_0^* \end{array} \right\}$$

4 distinct coeff. $\Rightarrow X_0$ is real

⊗ If TP is odd, then X_0 is real,

$$X_k = X_{N-k}^*$$

⊗ Total distinct Coeff. = $\frac{(N+1)}{2}$

⊗ ⊗ If N is even, then X_0 is also real &

$$X_0 = X_0^* = X_6^*$$

$$X_1 = X_1^* = X_5^*$$

$$X_2 = X_2^* = X_4^*$$

$$X_3 = X_3^* = X_3^*$$

$$X_4 = X_4^* = X_2^*$$

$$X_5 = X_5^* = X_1^*$$

$$X_6 = X_6^* = X_0^*$$

$\Rightarrow X_0$ is real

$\Rightarrow X_3$ is real

$\therefore$ For even TP ;

* $X_{\frac{N}{2}}$ & X_0 are real

* $\frac{N}{2}$ distinct Fourier Series Coeff.